

KOMATSU AB, Japan

WMO: 477040

Lat: 36.394N

Lon: 136.407E

Elev: 32

StdP: 14.68

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.5	30.2	17.8	13.5	37.8	20.8	15.6	37.3	33.8	38.4	30.1	37.6	6.2	120	0.450

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.3	91.8	76.1	89.7	76.0	87.7	75.7	78.9	87.3	77.8	85.9	76.8	84.6	12.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.7	139.4	83.5	75.3	132.9	82.5	73.8	126.2	81.6	42.4	87.5	41.3	87.0	40.1	84.8	86.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.6	22.0	19.1	DB	24.4	97.2	3.0	2.3	22.2	98.8	20.4	100.1	18.7	101.4	16.5	103.0	
			WB	22.9	81.1	2.6	2.3	21.0	82.8	19.4	84.1	18.0	85.4	16.0	87.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.6	38.7	39.2	44.8	53.8	63.1	70.1	78.3	80.8	73.2	63.1	52.5	43.5	
	DBStd	15.35	4.29	4.95	5.87	6.43	5.10	4.46	4.56	3.92	5.10	5.23	5.52	5.12	
	HDD50	1117	350	306	186	32	0	0	0	0	0	0	34	209	
	HDD65	3750	815	722	628	340	96	7	0	0	3	99	375	665	
	CDD50	4239	1	3	23	146	408	604	877	956	697	405	110	9	
	CDD65	1396	0	0	0	4	39	161	412	491	249	39	1	0	
	CDH74	11410	0	0	1	37	227	688	3586	5121	1593	154	3	0	
	CDH80	3813	0	0	0	5	28	126	1218	2003	411	22	0	0	
Wind	WSAvg	7.9	9.8	9.0	8.8	8.2	7.4	6.4	6.7	6.6	6.9	7.1	8.1	9.6	
Precipitation	PrecAvg	89.3	10.7	6.5	6.1	5.2	5.6	6.2	9.0	5.9	8.1	6.1	8.3	11.5	
	PrecMax	107.0	18.0	10.1	9.1	8.7	14.0	11.1	22.7	14.9	13.9	9.7	16.3	25.4	
	PrecMin	60.7	3.9	2.1	2.8	1.9	2.4	1.8	2.1	1.1	2.7	3.2	4.8	4.9	
	PrecStd	10.7	3.4	2.0	1.7	1.9	2.9	2.3	5.1	3.3	3.8	1.7	2.2	4.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.5	64.2	69.9	78.8	83.8	87.8	95.0	96.6	91.7	82.8	73.0	64.3	
		MCWB	49.8	53.4	53.8	60.6	64.7	72.3	75.6	76.7	73.9	68.7	60.6	53.9	
	2%	DB	52.1	55.8	64.5	73.3	79.0	83.8	91.4	93.2	87.6	77.4	68.4	58.9	
		MCWB	44.4	47.4	51.2	57.6	62.2	70.3	75.9	76.6	73.9	66.4	58.5	50.8	
	5%	DB	48.6	51.8	60.4	69.6	76.7	80.4	89.2	91.1	84.4	75.1	66.0	55.4	
		MCWB	42.6	44.3	49.8	56.3	62.2	69.7	75.8	76.6	73.3	64.9	57.1	48.9	
	10%	DB	46.4	48.2	55.7	66.0	73.4	77.4	86.2	88.2	82.1	72.0	62.7	52.2	
		MCWB	41.4	42.4	47.7	54.6	61.1	68.5	74.9	76.5	72.0	62.9	55.5	46.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.1	54.7	56.8	63.9	68.7	74.9	79.5	80.5	77.8	71.7	63.3	56.4
			MCDB	56.7	63.1	64.2	72.5	77.6	82.1	88.0	88.3	86.0	79.2	68.3	62.0
2%		WB	46.1	49.4	53.5	60.1	65.9	72.9	78.2	79.5	76.0	69.0	60.6	52.2	
		MCDB	50.1	54.8	61.5	68.6	74.0	78.9	86.3	87.6	83.1	74.8	66.3	57.1	
5%		WB	43.4	45.5	51.0	57.9	64.2	71.7	77.0	78.6	74.6	66.9	58.5	49.7	
		MCDB	47.5	50.0	57.6	66.3	72.3	77.8	84.9	86.9	81.3	72.8	64.3	54.2	
10%		WB	41.3	42.5	48.7	55.8	62.7	70.2	76.1	77.6	73.1	64.9	56.3	47.4	
		MCDB	45.6	47.2	55.3	63.3	70.1	75.8	83.6	85.7	79.9	70.9	61.7	52.0	
Mean Daily Temperature Range	5% DB	MDBR	11.1	12.6	15.7	16.9	15.7	12.1	11.6	13.3	13.4	15.3	15.1	12.7	
		MCDBR	16.2	19.9	23.8	23.6	21.5	16.5	15.9	16.5	16.1	17.8	19.9	17.8	
		MCWBR	11.2	13.8	14.7	12.1	10.2	7.5	6.2	6.7	7.4	9.3	12.2	12.2	
	5% WB	MCDBR	15.0	18.0	19.5	18.6	16.6	12.1	13.2	13.8	13.7	14.6	16.5	16.0	
		MCWBR	11.3	13.8	13.7	10.9	9.4	6.3	6.0	6.6	7.1	8.7	11.5	11.9	
Clear-Sky Solar Irradiance	taub	0.439	0.468	0.533	0.556	0.556	0.575	0.568	0.541	0.489	0.456	0.440	0.440		
	taud	1.951	1.897	1.802	1.812	1.874	1.908	1.988	2.067	2.166	2.141	2.092	1.985		
	Ebn at Noon	223	235	232	236	239	233	234	238	244	238	224	213		
	Edh at Noon	45	53	64	67	64	62	56	51	44	41	39	41		
All-Sky Solar Radiation	RadAvg	506	753	1145	1466	1659	1592	1549	1623	1248	992	657	446		
	RadStd	47	75	97	121	181	130	226	172	100	80	57	59		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (27 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page